Lesson 2.1 — Quadratic Functions in Standard Form

Big idea: Standard form $y = ax^2 + bx + c$ hands you the y-intercept and the axis of symmetry for free — if you know where to look.

Quick review

Standard form: $y = ax^2 + bx + c$. The graph is a parabola.
Direction: opens UP if $a > 0$, DOWN if $a < 0$. Larger $|a|$ = narrower.
Y-intercept: c (the constant term).
Axis of symmetry / vertex: $x = –b/(2a)$. Plug that x back in to get the vertex y-value.

Worked example

Worked example. $y = x^2 – 6x + 5$. Find direction, y-intercept, axis of symmetry, vertex, and zeros.

$a = 1 > 0 \rightarrow$ opens up. y-intercept: 5. Axis: $x = 6/2 = 3$. Vertex: $(3, 9 – 18 + 5) = (3, –4)$. Zeros: factor $x^2 – 6x + 5 = (x – 1)(x – 5) \rightarrow x = 1, 5$.

Warm-ups

1. For $y = 2x^2 – 4x + 1$, state the y-intercept.
2. For $y = –x^2 + 3x – 5$, does the parabola open up or down?
3. Find the axis of symmetry for $y = x^2 – 8x + 12$.

Practice

1. Find the vertex of $y = 2x^2 – 12x + 7$.
2. Find the zeros of $y = x^2 + 5x – 14$ by factoring.
3. Sketch (describe) $y = –0.5x^2 + 4x – 6$: direction, vertex, y-intercept.

Challenge

1. A parabola has y-intercept 7, vertex (2, –5). Find a, b, c in standard form.
2. For what values of k does $y = x^2 + kx + 9$ have its vertex on the x-axis?

Answer key — Lesson 2.1

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. y-intercept = 1.
2. Down ($a = -1 < 0$).
3. $x = 8/2 = 4$.

Practice

1. $x = 12/(2\cdot 2) = 3$. $y = 2(9) - 36 + 7 = -11$. Vertex: $(3, -11)$.
2. $(x + 7)(x - 2) = 0 \Rightarrow x = -7, 2$.
3. Opens down ($a < 0$). Axis: $x = -4/(-1) = 4$. Vertex: $(4, -0.5(16) + 16 - 6) = (4, 2)$. y-intercept: -6.

Challenge

1. $y = a(x - 2)^2 - 5$; plug $(0, 7)$: $7 = 4a - 5 \Rightarrow a = 3$. Expand: $y = 3x^2 - 12x + 7$.
2. Vertex on x-axis means discriminant = 0: $k^2 - 4(1)(9) = 0 \Rightarrow k^2 = 36 \Rightarrow k = \pm 6$.